Research Idea: eso_137-001

ALMA Thesis Planner (automated pipeline)

Kinematic structure of molecular clumps in the ram-pressure-stripped tail of ESO 137-001

Using the pipeline-calibrated CO(1-0) mosaic cube and image products already delivered for project 2013.1.01364.S (public since 2016-07-14, no re-observation needed), the student will characterize the kinematics of discrete molecular gas along the X-ray/Hα tail versus the disk, testing whether stripped-tail gas behaves as a dynamically distinct, in-situ-formed population rather than gas passively dragged out of the disk.

Concrete, bounded steps:

1. Load the QA2 cube (frequency setup spans 98.6–100.5, 100.6–102.5, 110.8–112.7, and 113.3–113.8 GHz; CO(1-0) rest frequency 115.27 GHz falls within this coverage once redshifted by the cluster's recession velocity — the data description does not state the assumed redshift, so the student must confirm the line's precise observed frequency against the delivered cube/QA2 documentation before analysis rather than assume a value) in CASA and produce moment 0, moment 1, and moment 2 maps at the native 7.892" angular resolution and 0.643 km/s velocity resolution.

2. Inspect the delivered mosaic footprint directly (image header/primary-beam coverage) to determine the actual spatial extent of usable tail coverage, rather than assuming the abstract's 80 kpc (X-ray tail) or 40 kpc (APEX single-dish CO detection) figures describe the ALMA field of view.

3. Flag missing-flux risk as a qualitative, data-description-only diagnostic: the observations table lists only a single 12m-array member OUS, with no ACA or total-power component. Compute the maximum recoverable angular scale (MRS) implied by this array configuration from the delivered measurement-set/QA2 documentation (e.g., via the shortest baseline in the calibrated data, or CASA's reported MRS for the imaging run) and compare it directly to the angular size of the tail structures being analyzed (converting kpc scales to arcsec at the cluster distance). If tail structures approach or exceed the MRS, report this explicitly as a known risk of resolved-out diffuse flux affecting any tail gas mass or morphology measurement — without invoking any specific external "APEX total" flux value, since no such number appears in this data description or in any archive-linked publication (none recorded). This keeps the diagnostic entirely within the delivered data and its own imaging metadata.

4. Use a simple clump-finding algorithm (e.g., a 2D/3D dendrogram or SExtractor-style thresholding on moment-0) to identify discrete CO condensations in the disk versus in the tail, within whatever extent step 2 confirms is covered and usably sensitive. Define the minimum viable deliverable up front: if fewer than ~5 independent resolution elements along the tail reach S/N > 3 in integrated intensity, individual clump kinematics will not be pursued; instead the thesis reports a tail-integrated (binned) upper limit on molecular gas mass and velocity dispersion, explicitly framed as a non-detection result with quantified limits rather than an open-ended fallback.

5. For each detected clump (or tail-integrated bin, per step 4's branch), measure centroid velocity, integrated flux (→ molecular gas mass via a CO-to-H2 conversion factor), and velocity dispersion. State explicitly that the adopted α_CO is a significant systematic uncertainty for this target: ram-pressure-stripped tail gas may differ in metallicity and excitation from disk gas, and α_CO calibration for stripped/in-situ-formed tail material is actively debated in this literature. Report tail gas masses with this systematic caveated separately from statistical uncertainty, and optionally show results under 2–3 literature-motivated α_CO choices as a sensitivity check rather than a single adopted value.

6. Compare the velocity field and dispersion structure of tail gas to disk clumps: does the tail show a smooth velocity gradient consistent with simple ballistic stripping, or more turbulent/decoupled structure consistent with local condensation out of mixed ISM–ICM gas? Given the modest number of independent resolution elements likely available, frame this as a descriptive kinematic characterization (or, per step 4, a bounded non-detection with limits) rather than a statistically powered population test.

This stays within already-public, already-imaged data products (no proprietary or unreleased data, no external catalogs or unnamed papers introduced), requires only moment-map generation, footprint/MRS-based missing-flux verification computed from the delivered data itself, a clump-finding pass with a pre-defined minimum deliverable, and a kinematic comparison with explicit systematic-uncertainty reporting — not the full multiwavelength heating/cooling, star-formation-efficiency, and turbulence study of the original proposal. It is tractable within CASA plus standard clump-finding/analysis tools and yields a clear, honestly-scoped, well-bounded result regardless of whether tail clumps are individually resolved.